TitrateRL: IMRaD Outline, Deployment, Repository Structure, Class Diagram and MVP of a Simulation-Based Reinforcement Learning Framework for Basal Insulin Titration

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***Abstract*—TitrateRL is an open, simulation-based framework in which reinforcement learning (RL) agents learn to titrate a once-daily basal insulin dose for virtual adults with Type 2 diabetes (T2D) and are compared with clinical titration algorithms. This report gives the one-page IMRaD outline of the study, the deployment map, the repository file structure and the class diagram of the implemented software, and the minimum viable product (MVP) defined with the MoSCoW method, together with the metrics used to accept the product and to answer the research questions. The software for every MVP feature is implemented and tested (84 automated tests); the simulator passes four of the five verification checks, cohort calibration is still open, and the agent experiments have not been run yet. All work is *in silico* and not intended for clinical use. Code: https://github.com/Nodirjn/TitrateRL.**

***Index Terms*—reinforcement learning, Type 2 diabetes, basal insulin titration, Gymnasium, MVP, MoSCoW**

I. IMRaD Outline

A. Introduction

- **Context.** 589 million adults live with diabetes, over 90% with T2D. Basal insulin (e.g., glargine) is started at 10 U/day and titrated by fasting glucose; at +2 U every 3 days reaching 50 U takes 60 days, and 81% of patients remain above their HbA1c target after six months.
- **Gap.** Clinical algorithms use one rule for everyone; 24 of 29 RL glucose-control studies address Type 1 diabetes; no open, RL-ready environment for daily T2D basal titration exists.
- **Aim.** Build an open benchmark and test whether a learned, shielded policy titrates better than clinical rules. **RQ1**: effect of the reward (R1 time in range, R2 risk index, R3 TIR with hypoglycemia penalties). **RQ2**: $\geq$5 pp more time in range than AT.LANTUS, Treat-to-Target and INSIGHT with non-inferior hypoglycemia. **RQ3**: dependence of the benefit on hidden insulin sensitivity.

B. Methods

- **Simulator.** Dalla Man T2D meal model (15 ODE states, RK4, 1-min step, Numba) with endogenous secretion, two-compartment glargine absorption, meals, CGM and fingerstick noise.
- **Cohort.** 150 virtual adults (fasting glucose, weight, insulin-sensitivity and β-cell factors), split 90/30/30 by insulin-sensitivity tertile; calibrated against the Treat-to-Target trial.
- **Decision problem.** Partially observable MDP, one step per day, 168-day episodes; observation = 3-day stack of 11 features; action $\in \{-4, -2, 0, +2, +4\}$ U; weekly rate limit 8 U, dose ≤ 1 U/kg.
- **Agents and baselines.** PPO (primary) and DQN (Stable-Baselines3), 10^6 days per run, 5 seeds; rule-based safety shield; baselines B0 (fixed), B1–B3 (clinical), B4 (tuned).
- **Evaluation.** 30 unseen patients $\times$ 10 scenarios with common random numbers; noise-free endpoints; one-sided Wilcoxon with Holm correction, Hodges–Lehmann estimate, bootstrap CIs, Spearman correlation.

C. Results (status and expected outputs)

- **Achieved.** All MVP software implemented; 84 tests pass; one 168-day episode runs in $\approx$0.15 s; PPO trains at $\approx$690 steps/s ($\approx$25 min per 10^6 steps).
- **Verification.** V1 (integrator, equilibrium, dose–response, identical right-hand side to `simglucose`) and V2 (flat glargine profile) pass; V4 calibration fails on fasting glucose (99.6 vs. 117 mg/dL in the trial) while the final dose matches (0.57 vs. 0.48 U/kg).
- **Preliminary software check** (validation patients, one scenario): clinical rules reach a median maintenance TIR of 60–63% vs. 39% without titration.
- **Expected.** Tables of TIR/TBR per reward (RQ1), H2 decision table vs. B1–B3 (RQ2), correlation and dose-behaviour plots by insulin-sensitivity tertile (RQ3).

D. Discussion (planned)

- Interpret efficacy–safety trade-offs of the rewards and the value of CGM features (PPO vs. FPG-only PPO) and of learning vs. tuning (vs. B4).
- **Limitations.** No glucagon (hypoglycemia interpreted comparatively); single-meal model extrapolated to 24 weeks; perfect adherence; calibration gap; sim-to-real gap.
- **Contribution.** Open benchmark `TitrateRL/T2D-v0`, pre-specified comparison, analysis of personalization. Not a medical device.

II. Deployment Map

Fig. 1 shows where each part of TitrateRL runs and how artifacts move. TitrateRL has no server back end and handles no personal data: the only deployed units are the Python package, the experiment runs and the Streamlit dashboard.

- **Repository.** https://github.com/Nodirjn/TitrateRL (MIT license).
- **Environments.** *Development*: each member's laptop. *CI*: GitHub-hosted Ubuntu runners, CPU-only PyTorch. *Experiments*: one multi-core CPU machine (no GPU needed; the networks are 2×64 MLPs). *Presentation*: local Streamlit server.
- **Reproducibility contract.** A run is identified by its YAML file, seed and commit hash, recorded in `run.json`; dependencies are pinned; data and models travel as release assets.
- **Safety.** The dashboard accepts only virtual-patient identifiers and labels every output as a simulation result.

III. Repository File Structure

The repository (Fig. 2) follows the `src` layout and mirrors the six architecture layers: configuration and data, simulation core, environment, decision, evaluation, and presentation. Experiment outputs (`data/`, `runs/`, `results/`, `reports/`) are generated and git-ignored.

Conventions. GitHub Flow with a protected `main`; Conventional Commits; one reviewer per pull request; CI runs ruff and pytest on Python 3.10–3.12; every experiment is a YAML file; milestones are tags `v0.1-env`, `v0.2-agents`, `v0.3-eval`, `v1.0-mvp`.

IV. Class Diagram

Fig. 3 shows the main classes. `T2DTitrationEnv` composes a `VirtualPatient`, which owns the parameter vector, the ODE state and three random streams (meals, CGM, fingerstick) for common random numbers. Rule-based baselines and RL agents share one abstract `Policy` interface, so the evaluator, the statistics module and the dashboard treat them identically. `SafetyShield` is a Gymnasium wrapper applied only to RL agents; the rate limit and dose bounds live in the environment and apply to every policy.

V. MVP Definition (MoSCoW) and Metrics

The MVP is the minimal set of working features that demonstrates the complete pipeline: virtual patients → training and shielded titration → statistical comparison → interactive visualization (Table I). *Must* items form the protected Core tier; *Should* items are dropped in reverse order if milestone M2 is more than one week late; *Could* items start only if M3 is on time.

A. Metrics

Two kinds of metrics are used: *product metrics* accept the software, and *research metrics* answer the research questions (Table II). Research endpoints are computed from noise-free plasma glucose over the 30 test patients $\times$ 10 scenarios; every patient contributes one paired value per comparison.

B. Current status against the MVP

All seven features are implemented, with 84 automated tests (including a cross-implementation test against `simglucose` and a smoke PPO training run). Remaining Must work is the cohort calibration (check V4: week-24 fasting glucose 99.6 mg/dL vs. 110–125 mg/dL; 56% vs. 25–50% of patients at target; final dose 0.57 U/kg within range), the 20 Core training runs ($\approx$15 CPU-hours) and the RQ1–RQ3 analyses, following milestones M1 (`v0.1-env`) to M4 (`v1.0-mvp`, Week 14).

Disclaimer

TitrateRL is in silico research software. It is not a medical device, and nothing in this report is medical advice.

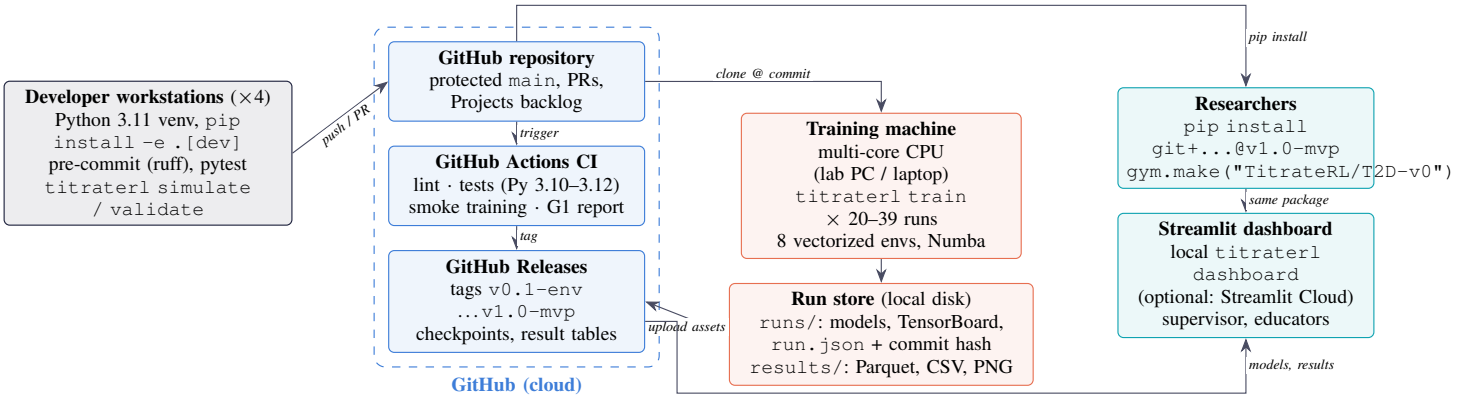


Fig. 1. Deployment map. Code flows from the four workstations through pull requests and CI to tagged releases; experiments run on a multi-core CPU machine from a pinned commit; trained models and result tables are published as release assets (never committed to git) and consumed by the dashboard and by researchers who install the package.

TABLE I
MVP PRIORITIZATION WITH THE MOSCOW METHOD

Priority	Feature	Acceptance criterion	Status
Must	F1 T2D simulator and calibrated 150-patient cohort	generated reproducibly from one seed; checks V1–V4 pass	implemented; V4 open
Must	F2 Gymnasium environment `TitrateRL/T2D-v0`	passes `check_env`; 168-day episode <2 s; rate-limit and shield tests pass	done (0.15 s)
Must	F3 Trained agents: PPO with R1–R3 (shield off) and shielded PPO with R*, 5 seeds each	checkpoints and learning curves stored; selection on validation patients	pipeline done; runs Wk 7–10
Must	F4 Rule-based policies B0–B3 behind the `Policy` interface	unit tests reproduce the published titration steps	done
Must	F5 Evaluation pipeline	one command regenerates metric tables, statistical tests and figures	done
Must	F6 Interactive dashboard	select a test patient and $\leq$3 policies; traces, doses, metrics, disclaimer; 24-week re-run <5 s; CSV export	done
Must	F7 Installable package	`pip install`; environment registered; documented API and example notebook	done
Must	RQ1–RQ3 analyses	H1–H3 decision rules applied to test-cohort data	Wk 10–13
Should	DQN agent; FPG-only PPO; λ grid for R3; simulator-tuned rule B4; robustness scenario set	completed and reported as exploratory	code ready
Could	Oracle per-patient fine-tuning; 2013 hypoglycemia model extensions; LSTM policy; SAC	started only if M3 is on time	—
Won't (this semester)	Real patient data or clinical deployment; Type 1 diabetes, pumps and bolus insulin; exercise, illness, missed injections; offline RL from EHRs; mobile app	out of scope; stated in README and model card	—

```
titraterl/
├─.github/ — CI workflow, PR and issue templates
├─configs/
│ ├─cohort.yaml
│ ├─train/ — ppo_r1/r2/r3, ppo_rstar_shield, dqn, fpg, smoke
│ └─eval/ — test.yaml, baselines_quick.yaml
├─src/titraterl/
│ ├─simulation/ — parameters, model (ODE, RK4), scenario, sensors, patient
│ ├─cohort/ — generator, split, Parquet I/O
│ ├─envs/ — titration_env, observation, rewards, wrappers
│ ├─policies/ — base, rules (B0–B4), sb3
│ ├─training/ — train, callbacks
│ ├─metrics/ — glycemic, episode
│ ├─evaluation/ — runner, statistics, logging, pipeline
│ ├─validation/ — checks V1–V5
│ └─config.py, cli.py
├─dashboard/app.py
├─tests/
│ ├─unit/ — model, cross-impl., env, shield, rules, ...
│ └─integration/ — pipeline, CLI, training, dashboard
├─docs/ — architecture, model, environment, evaluation, ...
├─examples/ — quickstart.py, quickstart.ipynb
└README.md, MODEL_CARD.md, pyproject.toml, Makefile, LICENSE
```

Fig. 2. Repository file structure (86 tracked files, ≈3 800 lines of package code and ≈800 lines of tests).

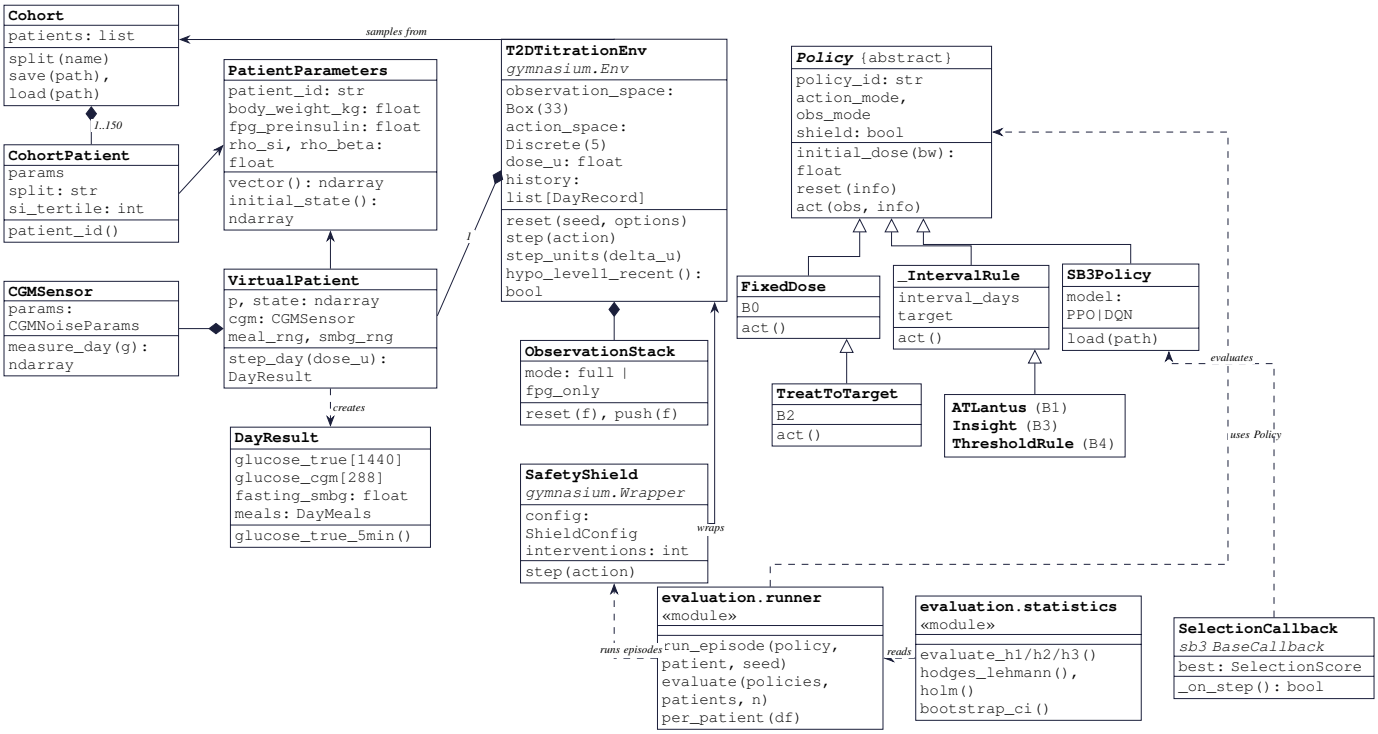


Fig. 3. Class diagram of the implemented TitrateRL package (simplified UML). Open triangles: inheritance; filled diamonds: composition; solid arrows: association; dashed arrows: dependency. B1, B3 and B4 inherit from `_IntervalRule`; B2 from `Policy`.

TABLE II
METRICS USED FOR THE MVP

Metric	Definition	Target / role
Product (acceptance)		
Test suite	unit + integration tests in CI	100% pass
Episode runtime	one 168-day episode	$<$2 s
Dashboard re-run	24-week simulation	$<$5 s
Simulator checks	V1–V4 (gate G1)	all pass by Wk 5
Reproducibility	same seed $\Rightarrow$ same trajectory	exact
Research (glycemic)		
TIR	% time 70–180 mg/dL, weeks 21–24	$>$70%; primary efficacy
$\text{TBR}_{<70}$, $\text{TBR}_{<54}$	% time below 70 / 54 mg/dL	$<$4%, $<$1%; primary safety
$\text{TAR}_{>180}$, $\text{TAR}_{>250}$	% time above 180 / 250	$<$25%, $<$5%
Mean, CV, GMI	mean, SD/mean, $3.31+0.02392\,\bar{G}$	CV $\leq$36%
LBGI, HBGI	Kovatchev risk indices	risk
Time-to-target	3 consecutive fasting values 80–130	titration speed
Nocturnal TBR	00:00–06:00	safety
Overbasalization	bedtime$-$morning $\geq$50; dose $>$0.5 U/kg	appropriateness
Dose burden	final U/kg; changes per week	burden
Shield rate	% days with intervention	safety
Statistical decision rules		
H2 superiority	Holm-adj. one-sided Wilcoxon	$p < 0.05$
H2 effect size	Hodges–Lehmann TIR gain	$\geq$5 pp
H2 non-inferiority	upper 95% CI of $\text{TBR}_{<70}$ diff.	$<$1 pp
H3	Spearman(ρ_{SI}, ΔTIR)	negative, $p < 0.05$